

What incremental steps should the Treasury consider towards additional transparency of secondary market transactions?

August 2026

Transparency: Since 2020, the Interagency Working Group on Treasury Market Surveillance has worked with FINRA to provide greater transparency in the secondary market as a means of enhancing the overall resilience of the Treasury market. What additional incremental steps should Treasury consider towards additional transparency of secondary market transactions?

FINRA began publishing Weekly Aggregate Statistics on Treasury market activity in 2020, which were converted to Daily Aggregate Statistics in 2023. Should the Daily Aggregate Statistics provide additional granularity about transaction volumes? For example, would it be beneficial for the Daily Aggregate Statistics to include additional breakdowns of activity within the current “Bills” category or on additional categories of off-the-run transactions for nominal coupons (not at the individual CUSIP level, but instead broad categories of near off-the-run vs. far off-the-run)?

FINRA began disseminating transaction-level data on transactions in on-the-run Treasury securities in 2024. How, if at all, has such publication affected Treasury market liquidity? Should the publication of transaction-level data be expanded or enhanced? For example, would it be beneficial to disseminate transaction-level data for bills or near off-the-runs? Would disseminating data one hour after a trade be of greater benefit than the current end-of-day publication of data?

How would such changes balance the generally accepted benefits of market transparency with potential concerns about the effects on liquidity provision? Are there other ways to further enhance transparency or address such concerns? Please elaborate.

Executive summary

What's changed:

- The Financial Industry Regulatory Authority (FINRA), through its Trade Reporting and Compliance Engine (TRACE), has gradually adopted several reporting and dissemination initiatives that have increased the transparency of transactions in the Treasury markets

What we found:

- Together with other factors, including the greater degree of electronification of Treasuries trading and the Treasury buyback program, this incremental transparency has contributed to a reduction in average transaction costs and greater market resilience
- While this improvement has occurred for both on-the-run (OTR) and off-the-run (OFR) Treasury securities, key differences remain across these markets, creating nuance in the impact of additional transparency initiatives

Suggested framework for evaluating additional enhancements:

- We lay out a framework for evaluating enhanced transparency proposals, with the following design principles:
 - Proposals should maintain market functioning, preserve anonymity, increase coverage and widen market participation
 - Implementation on this framework should consider:
 - If there are potential mitigants - such as delayed dissemination and broader bucketing,
 - Whether the preferred policy choices have broad market participant support, and
 - Whether changes can be implemented in a phased manner

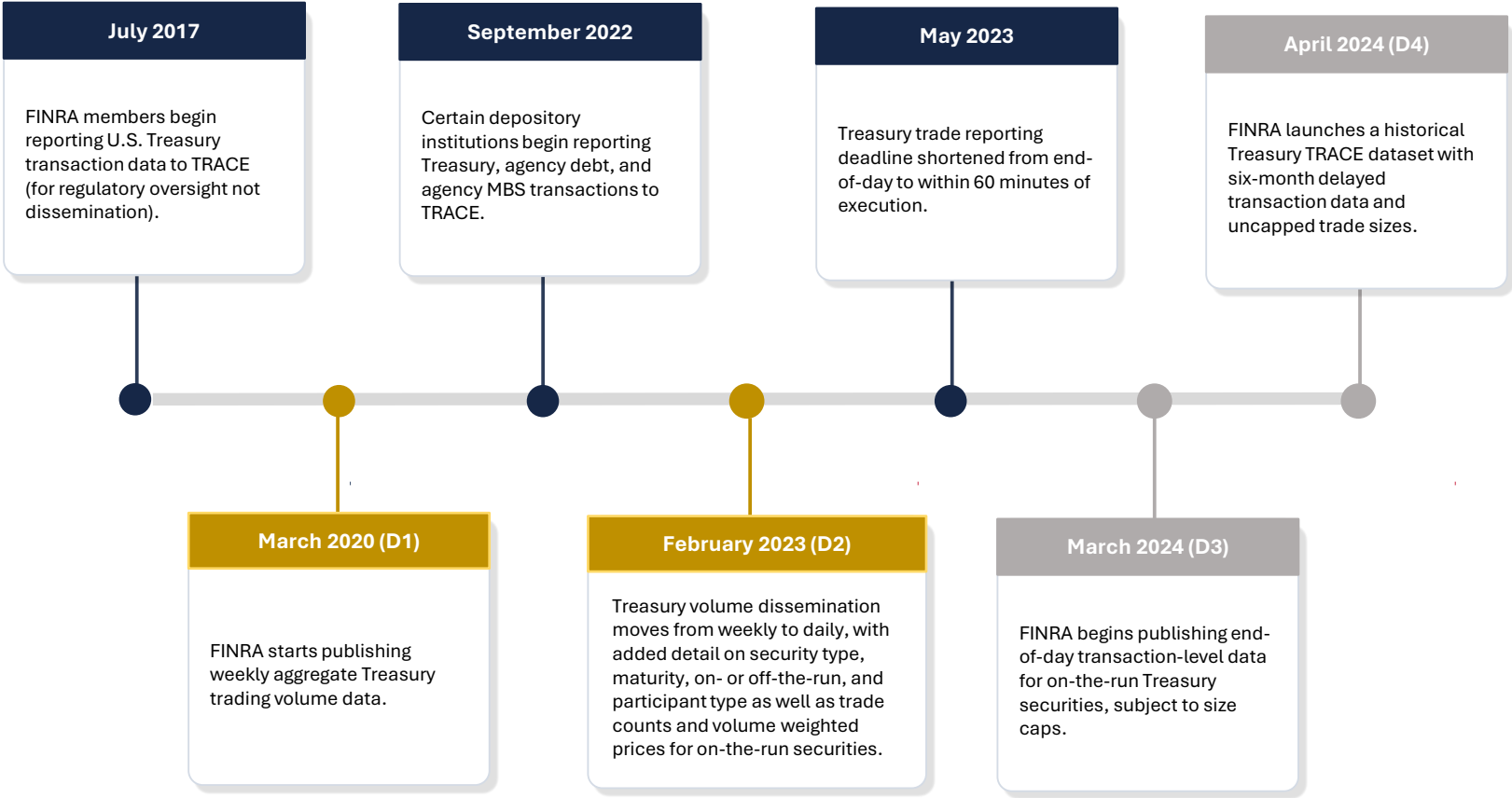
Potential next steps for consideration:

- Future transparency enhancements can be broadly bucketed as below:
 - Opportune next steps
 - **Enhanced dissemination for T- bills** - T-bills trading is less clustered around key events, CUSIPS are more fungible helping with dealer intermediation, and the transactions are more concentrated to primary markets
 - **Near real time disclosure for OTRs** – Potentially feasible due to sufficiently high trading volumes
 - Areas that warrant additional consideration
 - **Cheapest to deliver (CTD)** - Greater CTD transparency could provide insights into the large basis trade. Operational challenges need to be studied in greater detail and industry feedback established
 - Initiatives that may pose greater risk to market participant anonymity and/or dealer intermediation
 - **OFRs** - Due to their significantly lower turnover, more clustered trading patterns, higher costs to trade, and greater reliance on dealer intermediation, the impact of enhanced transparency can be materially different for OFRs than OTRs
 - **TIPS** - Holding of less than 2y maturity TIPS securities on dealer balance sheets has grown significantly and argues against increasing the granularity of maturity buckets for aggregate dissemination

Developments in improving Treasury transparency

Transparency-enhancing initiatives have gradually widened, moving from regulatory reporting toward broader public dissemination and more granular data

Blue denotes reporting initiatives; **gold** denotes aggregate dissemination initiatives; **gray** denotes transaction level dissemination initiative

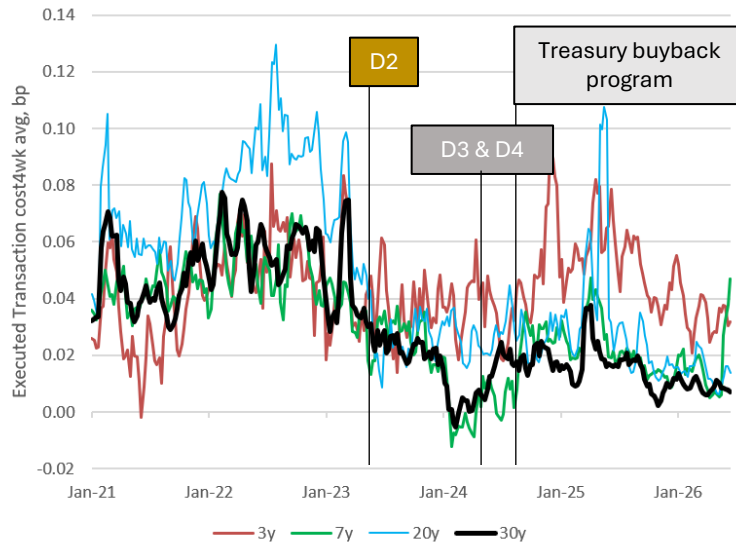


Key implication: Expanded transparency improves visibility into Treasury activity, but policy choices still need to balance transparency benefits against liquidity-provision concerns.

OTR executed average transaction costs have improved

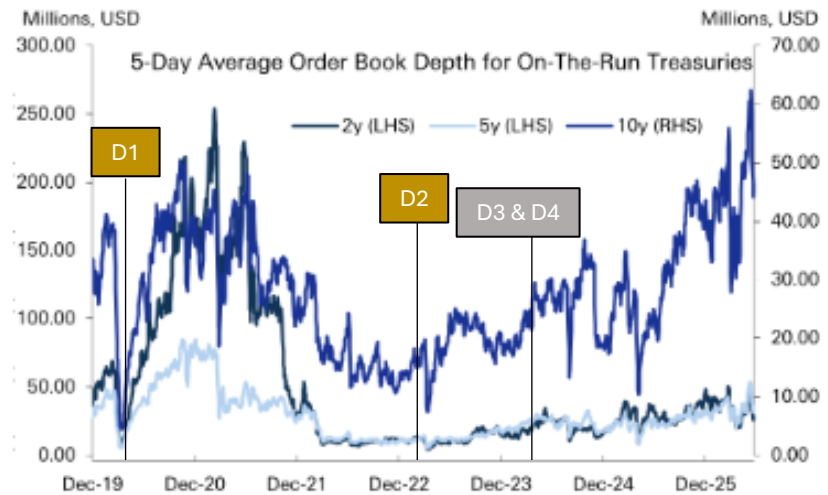
Transparency initiatives appear to have been absorbed without adversely affecting secondary market liquidity

OTR transaction costs have declined



Transaction cost calculated as the dv01 weighted average cost of executed transactions vs platform-wide Tradeweb mid in NY hours
Source: Tradeweb

Order depth for OTRs has increased



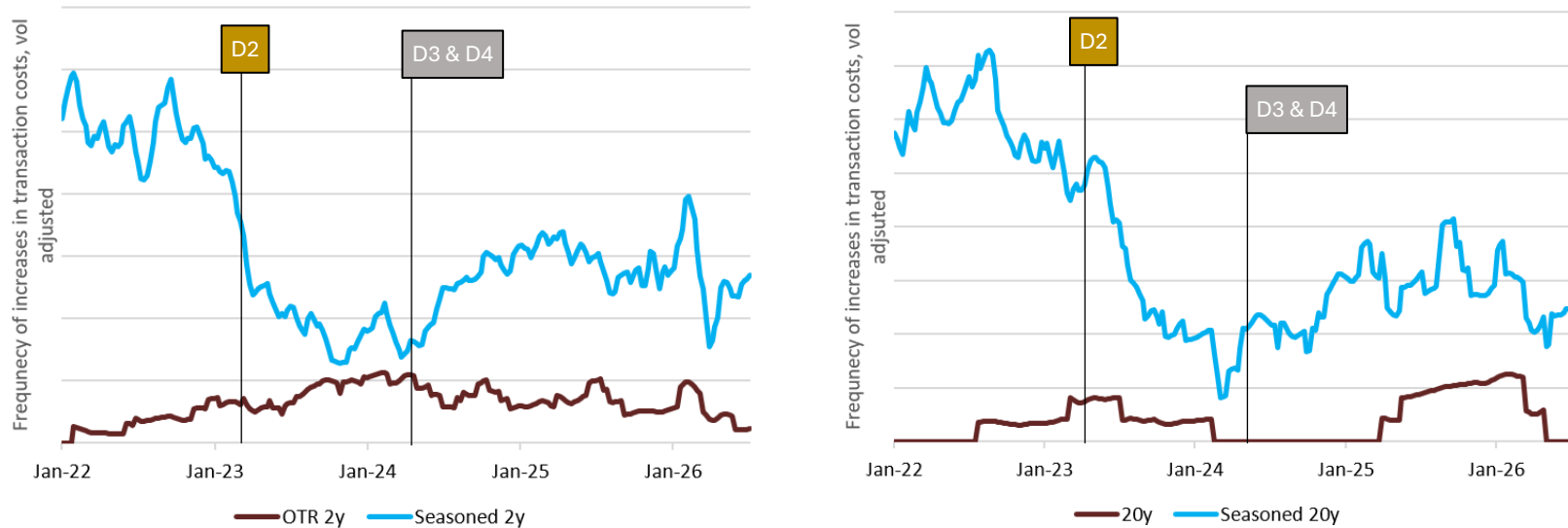
Order depth shown is 5day average of sum of bid and offer book sizes. Source: Deutsche Bank

- Transaction costs for OTR Treasuries have declined since 2022. This improvement has occurred in both average bid-offer spreads as well as consistency of those spreads (e.g., declining frequency of illiquidity episodes)
- Order book depth has strengthened over the past five years. Higher depth supports the ability to trade larger OTR sizes with less market impact
- Higher electronic trading and the Treasury buyback program have likely contributed

Reduced frequency of spikes in execution costs points to greater resilience

Even with the decline, this frequency remains 4x higher for OFR Treasuries than OTRs

For OTRs, weekly dv01 weighted executed transaction cost is higher than previous 3m average less frequently. While it has declined, this frequency remains significantly higher for OFRs than OTRs



Calculated as % of weeks, over a rolling 1y window, where T-costs are greater than previous 3m average by a threshold amount, normalized by 3m implied vol
Source: Presenter's calculations

- Transaction cost averages do not capture adequately the improvement in market resiliency
 - Resilience can be defined in terms of how frequently market participants face a rise in executed transaction costs during periods of increased volatility or market stress
- The frequency of “spikes” in OTRs execution costs has declined sharply over the past two years
- OTRs trade materially differently than OFRs
 - The frequency of OFR spikes has also reduced meaningfully over this period but still is 4x higher than OTRs

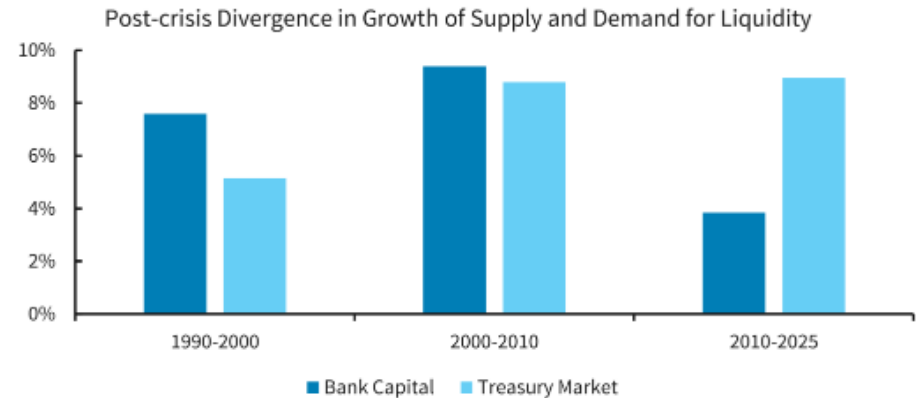
Changes to marketplace since the last proposal

Changes to key participant types, growth of the market, and policy are important factors to consider

Recent developments in the Treasury markets

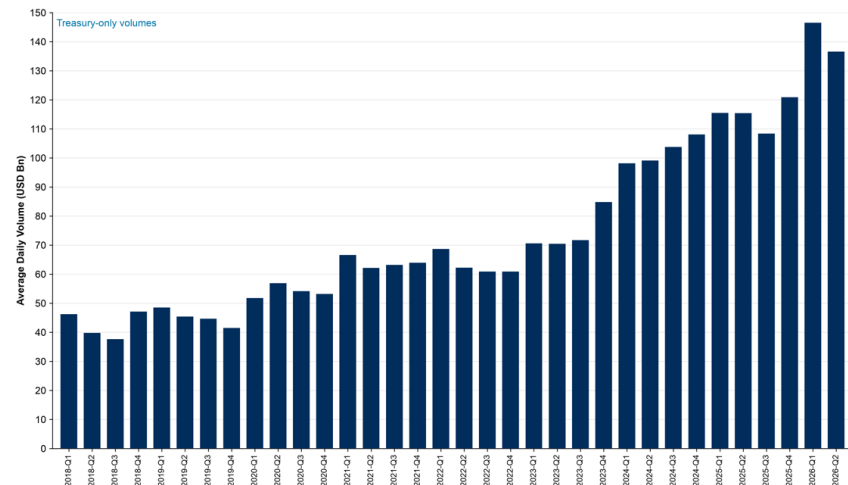
- **The market has grown meaningfully** - Growth of outstanding Treasuries has outpaced growth in bank capital
Impact: *The durability of dealer intervention during times of stress bears closer watching at sector level*
- **Central clearing to be implemented** - Expanded repo clearing to improve dealer balance sheet capacity
Impact: *Intermediation is expected to be positively impacted*
- **Treasury buyback program** – has improved the trading environment in seasoned securities
Impact: *Helped improve OFR liquidity on the margin*
- **Continued rise in electronic trading** – Significant acceleration in electronic trading volumes (~40% growth from 2018-2022 and ~100% from 2022-2026)
Impact: *Helped to increase transparency and create greater competition in RFQs*

Treasury supply has outpaced bank capital growth



Source: Barclays

Average daily US Treasuries volume transacted on Tradeweb



Source: Tradeweb

Framework for evaluating enhanced transparency proposals

Design Principles

Proposals for enhanced transparency could be evaluated along the following dimensions:

1. **Maintain market functioning:** Specifically, maintain intermediation for securities that rely heavily on dealer balance sheets for liquidity
2. **Preserve anonymity:** Dissemination of trading information should not allow for deduction of
 - Individual market participant identities, or
 - Their trading patterns, seasonality or sizes
3. **Increase coverage:** Make the set of Treasuries transactions in scope more comprehensive for reporting and/or dissemination
4. **Widen market participation:** Potentially expand market making/liquidity providers in specific market segments through reduced barriers to entry. For example, greater principal trading firm (PTF) participation in OFR sectors

Evaluation and Implementation Considerations

Evaluation and implementation on this rubric should consider:

- If risk-reward balance can be improved through potential mitigants, such as:
 - **Delayed dissemination:** Can this improve both intermediation and anonymity?
 - **Broader bucketing:** Near OFR/far OFR, or CTD identifiers
 - **Detail level:** Use of aggregate versus transaction-level data
- If the preferred policy choices have broad support among different market participants
- If it can be implemented in a phased manner, recognizing that initiatives may be difficult to unwind

Current FINRA TRACE dissemination

Past dissemination efforts have resulted in broadest availability of statistics for OTR securities

Level	End of Day details	Treasury Bills (T-bills) and FRNs	On-the-run Treasury Notes/Bonds (OTRs)	Off-the-run Treasury Notes/Bonds (OFRs)	TIPS
Aggregate Level	ATS & Interdealer trade count and volume	✓	✓	✓	✓
	Dealer to customer trade count and volume	✓	✓	✓	✓
	Total trade count and volume	✓	✓	✓	✓
	Weighted average price	✗	✓	✗	✗
	Sector detail	✗	✓	✓	✓
	OTR/OFR detail	✗			✓
Transaction Level	End of day released with capped size*	✗	✓	✗	✗
	6-month lagged with uncapped size*	✗	✓	✗	✗

Note: Data does not include certain transactions categories (e.g., foreign investor trades between non-FINRA counterparties)

* Data includes price, volume, time, buy/sell, counterparty type

Potential next areas for enhancement

Enhanced dissemination for T-bills and near real-time OTRs are the opportune next steps. Additional analysis warranted on operational feasibility and benefit for enhancement to CTDs

Potential “Next-step” Enhancement		Maintains Market Functioning	Preserves Anonymity	Increases Data Coverage	Widens Participation	Notes
Aggregate Level Dissemination						
T-bills	Increase granularity (maturity and OTR/OFR status) on EOD basis					Participants likely less sensitive to greater disclosure. Beneficial on balance
Off-the-runs	Separate near-OFR (1OFR and 2OFR) and far-OFR (>2OFR)					May impair anonymity due to concentrated trading profile. Dealer intermediation risks
TIPS	Increase granularity by maturity (<=2y and 2-5y segments)					High risk due to concentrated trading and dealer balance sheet dependence

Transaction Level Dissemination						
T-bills	Bring in line with current OTRs					Participants likely less sensitive to greater disclosure. Beneficial on balance
On-the-runs	Near real-time (e.g., 1 hour)					Lower risk due to sufficiently high trading volumes
Off-the-runs	Near OFRs					May impair anonymity. Dealer intermediation risks.
	Separate CTDs					Could provide meaningful information on basis trade, however operational complications due to changing CUSIPs



Beneficial effect



Adverse effect



Neutral or uncertain effect

Market participants have range of perspectives on impact of transparency

Summary of comments on Treasury's additional post-trade transparency RFI (2022)¹

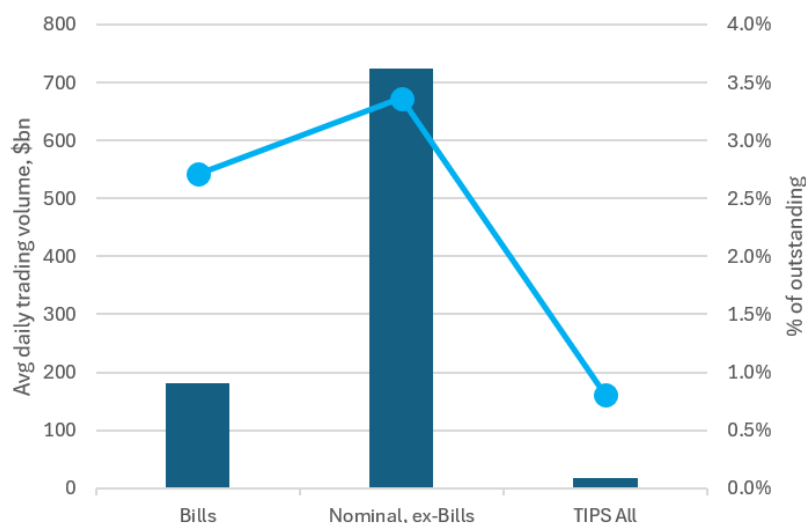
Issue	Principal Trading Firms	Buy-Side Firms	Sell-Side Firms
View on Transparency	Strongly supportive of transaction-level public reporting. Transparency improves competition, execution quality, and liquidity.	Support transparency as long as it protects anonymity, doesn't harm market functioning, and is consistently applied.	Generally skeptical of broad public dissemination and focused on liquidity risks.
Would Transparency Improve Liquidity?	Yes. Strong view that transparency broadens liquidity provision and reduces information asymmetries.	Yes, as long as it does not remove trading anonymity or disrupt dealer business models	Generally, no. Believe transparency could reduce market depth and dealer intermediation.
Treatment of Large Trades	Support block thresholds and delayed dissemination for very large trades.	Broad support for block thresholds, delays, and caps.	Strong support for robust block-trade protections and longer delays.
On-the-Run vs. Off-the-Run Treatment	Generally, support special treatment for less-liquid securities but favor eventual broad transparency.	Strong support for differentiated treatment by market segment.	Very strong support for differentiated treatment; view off-the-run market as particularly sensitive.
Phased Implementation	Generally, support phased rollout toward eventual comprehensive transparency.	Broad support for gradual implementation and reassessment.	Strong support for pilots, phased implementation, and extensive study before expansion.
Role of Regulatory Reporting	Support comprehensive reporting and public dissemination.	Strong support for comprehensive regulator access to transaction data.	Strong support for regulator access even where public dissemination is opposed.

Source: <https://www.federalregister.gov/documents/2022/06/27/2022-13540/notice-seeking-public-comment-on-additional-transparency-for-secondary-market-transactions-of>

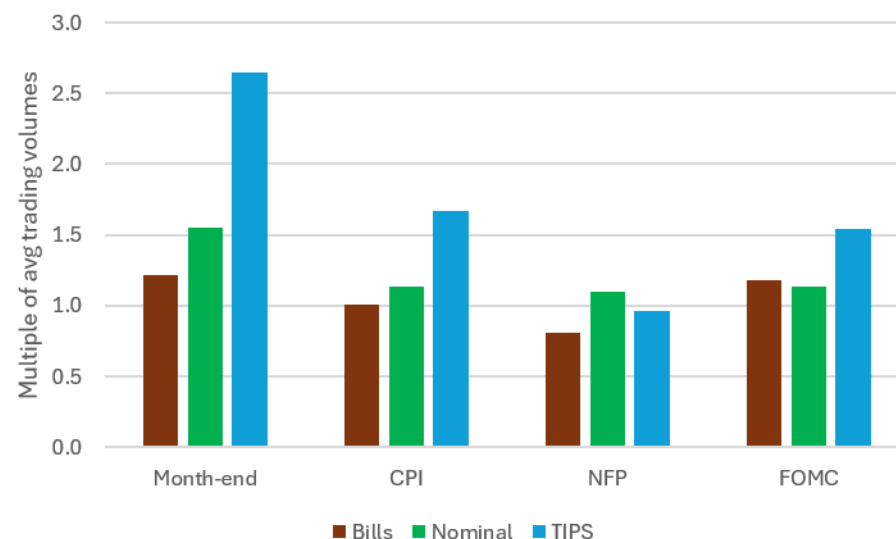
Nominals, T-bills and TIPS have materially different turnover and trading profile concentrations, making nuanced transparency rules necessary

Daily turnover is larger for nominals than TIPS (likely driven by OTRs)...

...in addition, trading volumes at month ends are multiples higher (especially for TIPS)



Source: FINRA, Bloomberg

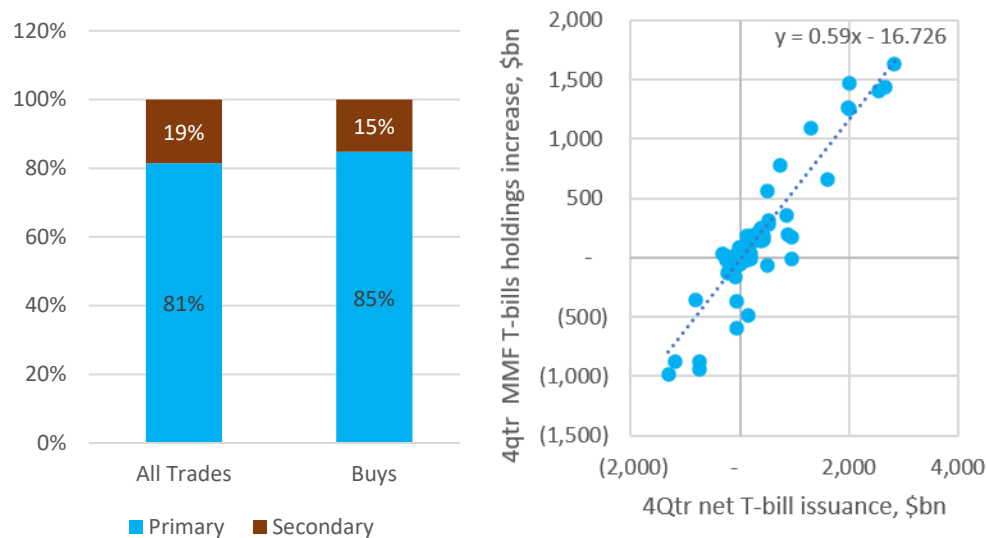


Treasury Sectors Exhibit Distinct Trading and Liquidity Profiles

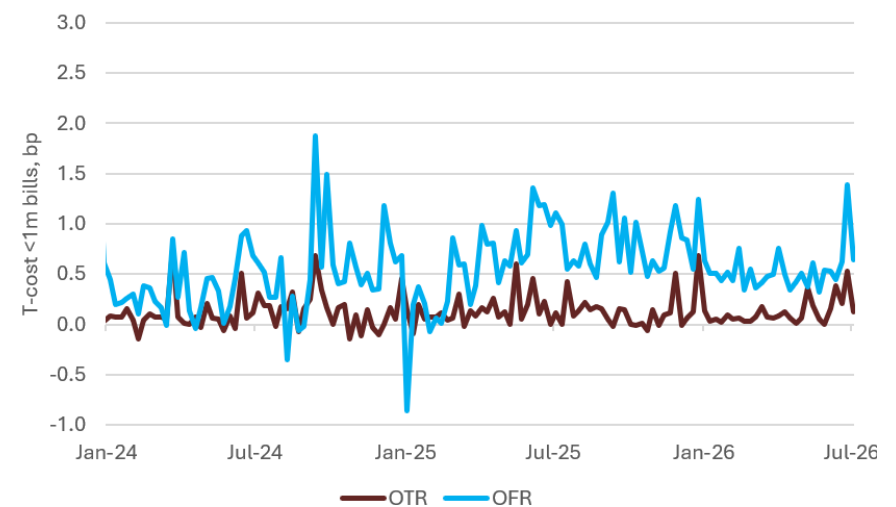
- Nominal Treasuries trade significantly more actively than other Treasury sectors, with daily turnover roughly 1.25x that of T-bills and 4x that of TIPS, as a share of amount outstanding
- Trading activity is distributed differently across Treasury sectors, both through the month and around key market events
- TIPS trading is heavily concentrated around month-end rebalances, with volumes averaging 2.7x normal daily levels, compared with 1.6x for nominal Treasuries and 1.2x for T-bills
- The greater concentration of TIPS trading around month end reflects the higher ownership by indexed portfolios
- TIPS trading is also more concentrated around major macro events, particularly CPI releases and FOMC meetings
- By contrast, T-bill trading volumes remain relatively stable throughout the month

Enhanced transparency on T-bill transactions likely to be beneficial on balance

Majority of money market funds' (MMF) T-bills purchases are in the primary market. MMFs have historically absorbed 60% of net supply.



Transaction cost differences between OTR/OFR T-bills are better behaved than for seasoned coupon Treasuries or TIPS



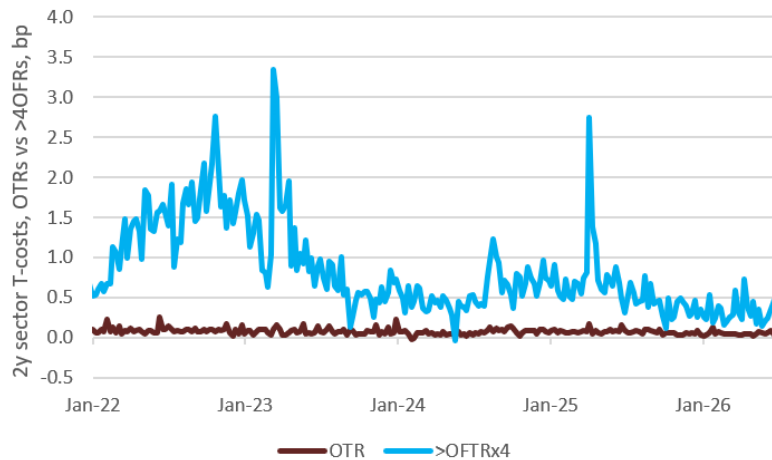
Source: Presenter's calculations, Federal Reserve, Tradeweb

- As Treasury issuance of T-bills increases, expanded transparency may benefit this market through increased coverage
- T-bills trading volumes average ~\$200-250bn a day, with turnover near the levels for nominal Treasuries
- T-bills trading activity is more evenly distributed over time and less concentrated around market events
 - Reducing concerns about anonymity or intermediation
- MMFs, which absorb ~60% of net T-bill issuance, execute ~80% of their activity in the primary market
- The short duration of T-bills reduces dealer warehousing risk
- High substitutability across T-bills CUSIPs enables greater flexibility in intermediation
- Trading costs for OFR T-bills appear more resilient than for seasoned coupon Treasuries or TIPS
- Through the monthly N-MFP process, MMF holdings are disclosed on at least a monthly basis, but some funds choose to disclose at a greater frequency
 - Likely indicative of lower sensitivity to greater incremental transaction dissemination

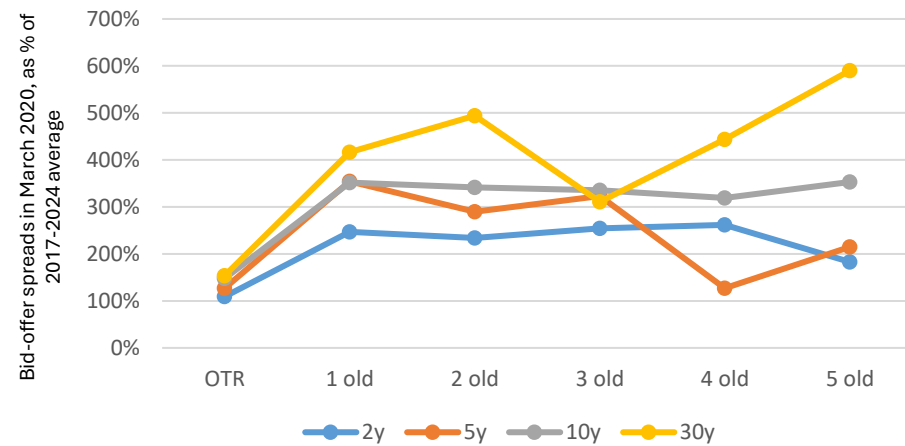
Dealer balance sheet constraints are visible in OFR trading costs

While these constraints have improved, and could further improve with more deregulation, fragility remains

OFR trading costs for seasoned Treasuries have improved but remains materially higher than OTRs



Transaction costs for OFR Treasuries increased significantly more than OTRs during the March 2020 episode



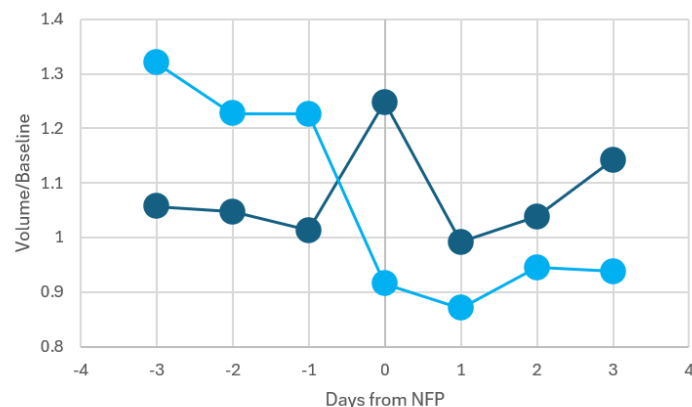
Source: FINRA, Tradeweb, Federal Reserve [Alain Chaboud et al]

Higher transaction cost for seasoned securities

- Based on dv01 weighted executed trades, cost to trade seasoned securities has improved materially since 2023
- However, it remains significantly higher than OTRs
- Likewise, the frequency of illiquidity episodes is also higher for seasoned securities
- Evidence that suggests that seasoned securities played an outsized role during illiquidity episode of March 2020

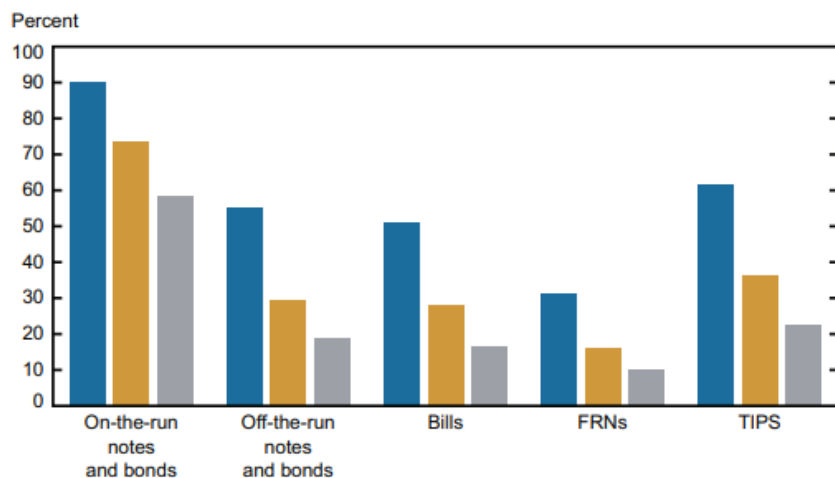
Longer dealer risk hedging timeline requires careful evaluation of increased disclosure proposals

OTR trading volume increases sharply on NFP days (relative to its average), while OFR volume declines marginally



Source: Presenter's calculation

% of dealer to customer flows that are offset by other customer flow within 15min, 1hr and daily



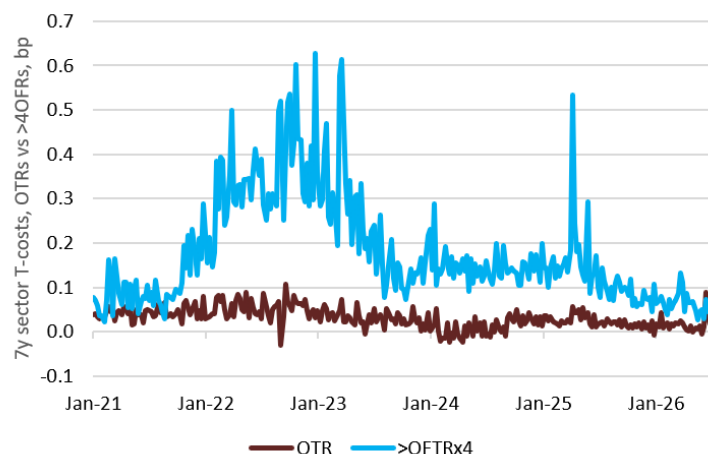
Source: https://www.newyorkfed.org/medialibrary/media/research/epr/2025/epr_2025_all-to-all_chaboud.pdf
[Alain Chaboud et al]

OTR and OFR have potential roles in buy side portfolio construction

- Off-the-run Treasuries are 98% of marketable Treasury debt outstanding. However, spike in volumes on key market dates such as Non-Farm Payroll (NFP) suggests that large risk transfer happens using OTRs on these days
- Buyers of off-the-run securities tend to be more “buy and hold” and use it for duration and key rate duration (KRD) management purposes
- Off-the-run trading tends to rely more heavily on dealer intermediation as there is less two-way order flow that allows for matching of buyers and sellers without dealers taking on inventory
- FINRA TRACE data suggests that purchases and sales of less liquid Treasuries at around the same time are relatively uncommon
 - Only ~20% of customer trading activity in off-the-run notes and bonds has offsetting activity in the same security within the same 15-minute interval, while this is 60% for OTRs

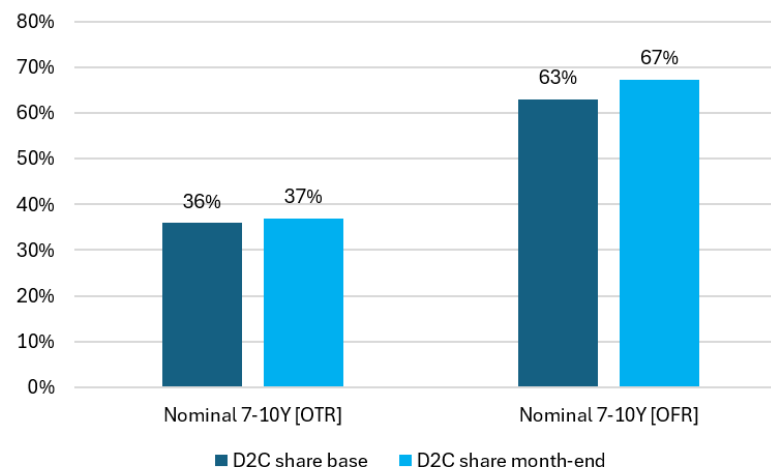
Greater dealer to customer (DTC) flow share for OFRs could cause greater transparency to adversely affect dealer intermediation

Transaction costs for >4old 7y sector Treasuries are materially higher than OTRs



Source: FINRA, Tradeweb

Two-thirds of OFR nominal transactions are DTC



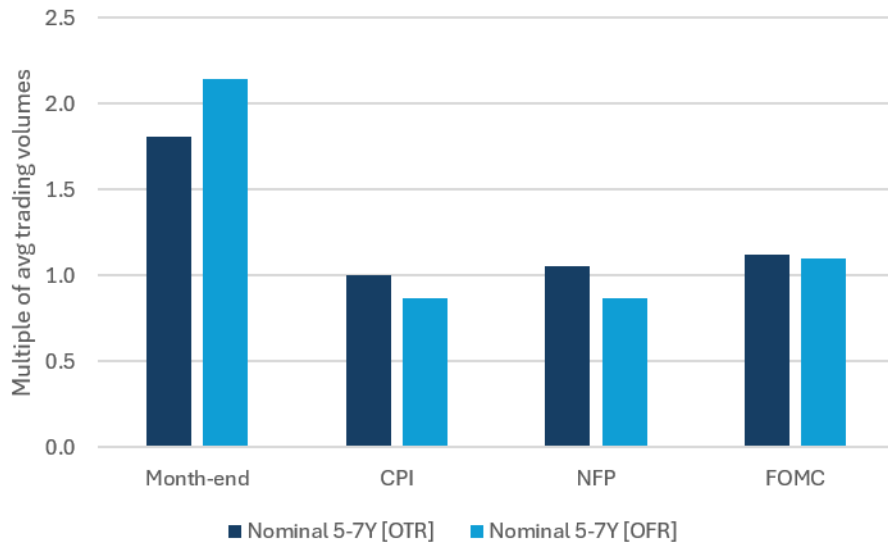
Source: FINRA

- Off-the-run (OFR) Treasuries, which rely on dealer intermediation to a greater degree, play an important role in month-end index rebalances
- While near OFRs (1 old or 2 old) have a lower cost than far OFRs, more granular disclosure, even with a delay, might present challenges for market participant anonymity:
 - Since the number of CUSIPs qualifying as near OFR would be small, aggregate disclosure could still be driven by large trades in select few names
 - Disclosure of near OFR securities could also, by default, make the reported statistics of far OFRs even more siloed
 - While delayed disclosure could mitigate some dealer-intermediation risks, OFR trading strategies are longer-lived and are often repeated systematically. This makes them vulnerable to reverse-engineering with delayed data

Key implication: Increased aggregate reporting that separates near-OFR from far-OFR creates challenges for participant anonymity. Delayed disclosure does not eliminate this risk.

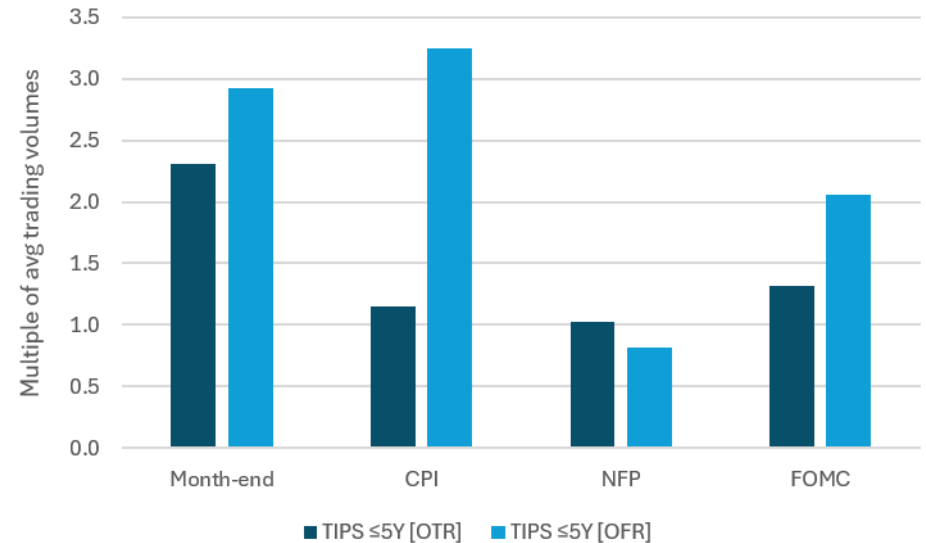
OFR trading coincides with predictable, periodic events, creating identifiable trading data, even with delayed release

OFR trading volumes are significantly higher on month ends but lower on data announcements



Source: FINRA

Increase in OFR TIPS volumes on month-end, CPI and FOMC days far outpaces that for OTR TIPS



Broad sector level data doesn't capture the full extent of concentration in off-the-run trading profiles

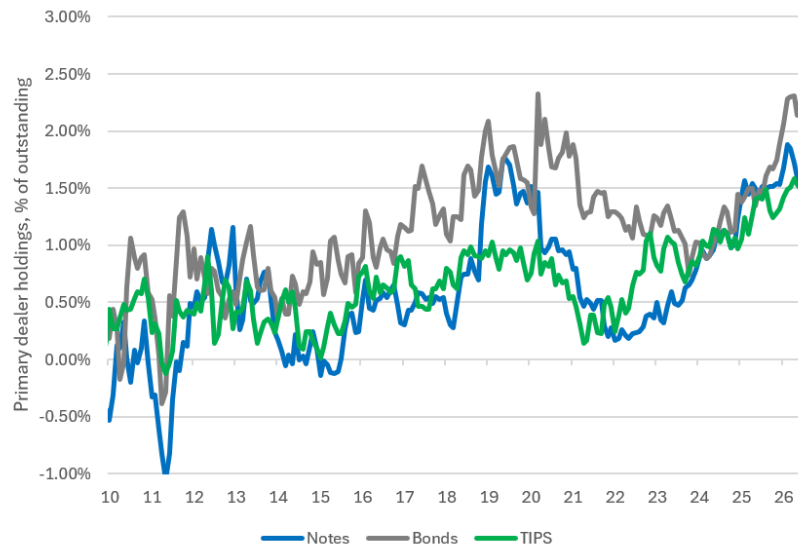
- While OTR trading volumes are 35% higher than their daily average on month end, OFR volumes are 145% higher in the <2y sector. Similar patterns occur across maturity profiles.
- OFR TIPS volumes are 2.9x on month-ends than their average and 3.25x higher on CPI days. These figures for OTR TIPS are 2.3x and 1.2x respectively
- This has implications for transparency disclosure requirements. A more concentrated OFR trading profile necessitates a careful evaluation of intermediation details

Source: FINRA

Primary dealers' holdings of Treasuries have increased

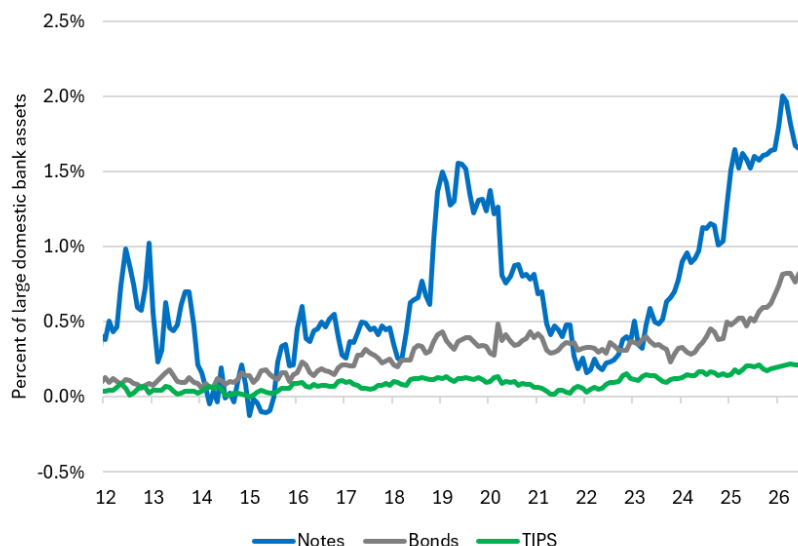
This increases the importance of dealer intermediation capacity, particularly for TIPS and nominal OFRs

Primary dealers' holdings of Treasuries have increased substantially, even as a percentage of outstanding ...



- With increased Treasury gross issuance, primary dealer holdings of Treasuries has increased markedly, more so for TIPS than nominals.
- This increase is not just outright, but is also as a percentage of outstanding
 - Nominal notes and bonds are back to their pre-pandemic peaks, having risen materially over the past 5 years
 - TIPS are about 50% higher than their pre-pandemic peak
- Nominals and TIPS OFRs are more heavily reliant on dealer intermediation, particularly during key market events such as month-ends

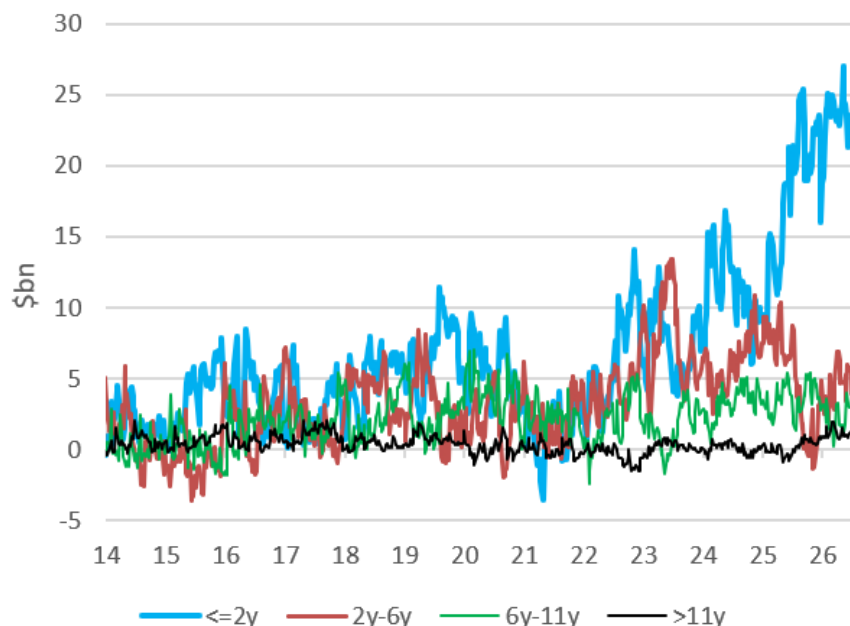
...particularly as a percentage of bank assets



- The rise in primary dealer holdings is also evident as a percentage of large domestic bank assets
- While hard to attribute to any single factor, regulatory changes have likely helped improve market conditions on the margin, which can be observed in the recent widening in front-end Secured Overnight Financing Rate (SOFR) swap spreads

Front end heavy primary dealer holdings argue against more granular TIPS disclosure – would impact intermediation and anonymity adversely

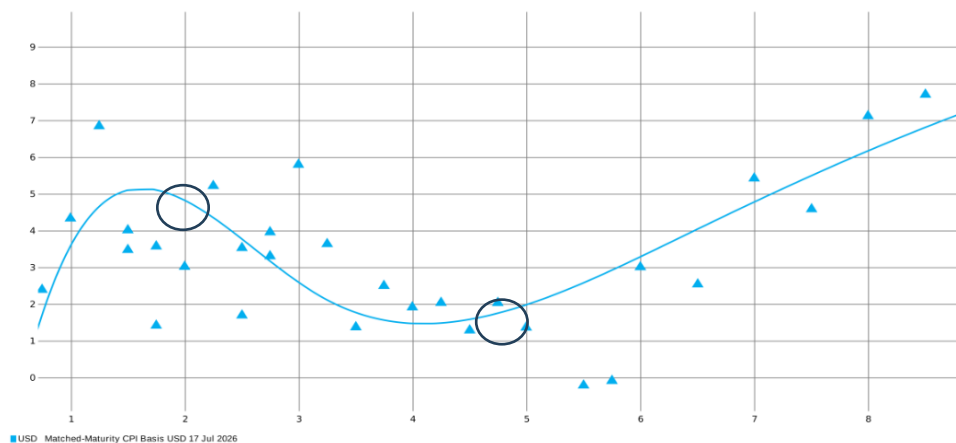
Primary dealer balance sheet holdings of <2y TIPS have increased substantially



Room to match nominal and TIPS maturity bucket distribution

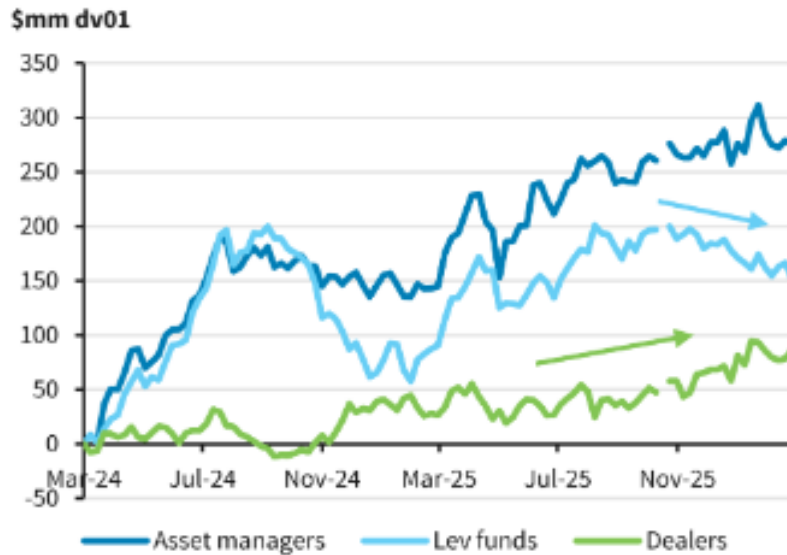
- Primary dealer balance sheet holdings of <2y TIPS have increased significantly over the past few years
- These securities have significantly greater volatility due to higher correlation with commodity prices and macroeconomic releases
- This likely results in <2y TIPS trading cheaper on IOTA than securities further out the curve
- Greater trading disclosure, even in aggregate, in the <2y sector could potentially adversely affect this asset class, which is more indexed than nominal Treasuries

Matched maturity IOTA spreads show 2y TIPS trade at a discount relative to 5y TIPS



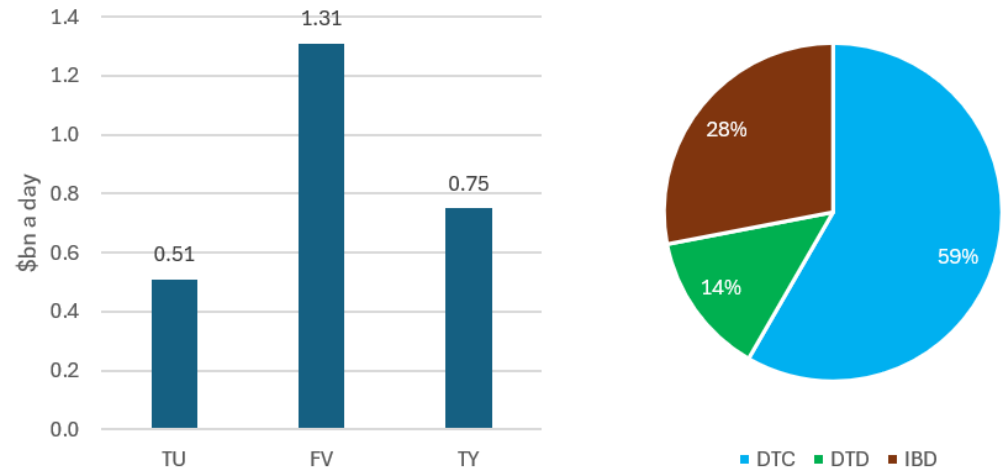
CTD status generates additional trading activity; more transparency might provide greater insight into the Treasury basis trades

Basis trade is shifting from levered funds to dealers



Source: Barclays, Federal Reserve

CTD securities have traded with materially higher volumes, ~40% of this increase is dealer-to-dealer or inter-broker dealer



Data from 2017-2024 Source: Federal Reserve

- Basis trade shifting to dealers from hedge funds
- CTD trading profile is closer to OTRs than it is to seasoned Treasuries
 - For example, between 2017 and 2024, fair value CTD traded \$1.3bn a day more than counterfactual
 - 60% of this increase is likely DTC (dealer to customer), while rest is dealer-to-dealer (DTD) or inter broker dealer (IBD)
- Given the growing interest in basis trade measurement,¹ the risk reward balance for CTD transactions disclosure might skew in favor of more dissemination
- This needs to be weighed against operational complexities such as CTD switching and co-CTDs
 - Over time, while this disclosure would refer to a changing set of CUSIPs, it might still be meaningful as a category

¹See link for additional information: <https://www.federalreserve.gov/econres/notes/feds-notes/quantifying-treasury-cash-futures-basis-trades-20240308.html>

Conclusion

- Treasury market transparency has increased meaningfully over the past few years. Together with other factors, including the greater degree of electronification of Treasuries trading and the Treasury buyback program, greater transparency has contributed to a reduction in average transaction costs and greater market resilience
- Greater resilience has been evident across both on-the-run (OTR) and off-the-run (OFR) Treasuries, but important structural differences between these markets suggest that additional transparency measures are unlikely to have uniform effects
- The most opportune next steps for additional transparency may lie in areas where risks to market participant anonymity and dealer intermediation are low, such as in expanded T-bills and near-real time OTR transaction level data dissemination
- Areas that may warrant additional analysis include trade-offs in expanding transparency in cheapest-to-deliver (CTD) transactions. While it may provide greater insight into the large basis trading positions, it may also be operationally complex
- A measured, targeted approach to transparency enhancements is likely to deliver the greatest market benefit while minimizing unintended consequences